MECHANICAL WORK, ENERGY AND POWER.

Introduction

The term 'work' as understood in everyday life has a different meaning in scientific sense. If a woman is carrying a load on her head and waiting for the arrival of a bus, she is not performing any work in the scientific sense. In this chapter, we shall look at the scientific meaning of the term '*work'* and different cases where work is said to be done. In addition to this, we shall also discuss the terms '*energy'*, types of mechanical energy. The term *'power'* will be defined and solved problems related to the topic will be considered.

Work

The physical meaning of the term work is entirely different from the meaning attached to it in everyday life. In everyday life, the 'work' is considered to be synonymous to 'labour', 'toil', 'effort' etc. In physics work is defined as;

Definition: *Work is said to be done by a force when the force produces a displacement in the body which it acts on in any direction except perpendicular to the direction of the force.*

For work to be done, the following two conditions must be fulfilled.

i. A force must be applied.

ii. The applied force must produce a displacement in any direction except perpendicular to the direction of the force.

Suppose a force **F** is applied on a body in such a way that the body suffers displacement **S** in the direction of the force. Then the work done is given by

$$W = \mathbf{F}.\mathbf{S} \qquad \textbf{5.1}$$

However, the displacement does not always take place in the direction of the force. Suppose a constant force **F** inclined at an angle θ applied to a body produces a displacement **S** as shown in Figure 5.1. Now the work done will be given by the dot product of force and the displacement.

$W =$ *component of force in the direction of displacement* **x** *the displacement.*

$$\therefore W = \mathbf{F}.\mathbf{S} = FS\cos\theta = (F\cos\theta)S \qquad \textbf{5.2}$$

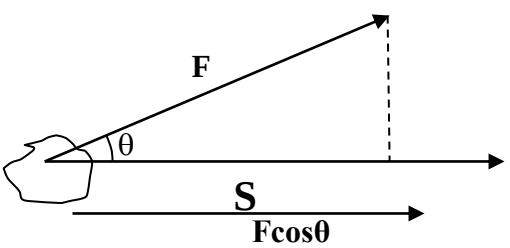


Figure 5.1 Force F acts on the object at an angle θ to the horizontal.

So, work is the product of the component of force in the direction of displacement and the magnitude of the displacement.

In terms of vector components, **F** and **S** may be written as:

$$\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k} \qquad \textbf{5.3}$$

$$\mathbf{S} = S_x\mathbf{i} + S_y\mathbf{j} + S_z\mathbf{k} \qquad \textbf{5.4}$$

$$\mathbf{F}.\mathbf{S} = \left(F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}\right)\left(S_x\mathbf{i} + S_y\mathbf{j} + S_z\mathbf{k}\right) \qquad \textbf{5.5}$$

$$= F_xS_x(\mathbf{i}.\mathbf{i}) + F_xS_y(\mathbf{i}.\mathbf{j}) + F_xS_z(\mathbf{i}.\mathbf{k}) + F_yS_x(\mathbf{j}.\mathbf{i}) + F_yS_y(\mathbf{j}.\mathbf{j}) + F_yS_z(\mathbf{j}.\mathbf{k}) + F_zS_x(\mathbf{k}.\mathbf{i}) + F_zS_y(\mathbf{k}.\mathbf{j}) + F_zS_z(\mathbf{k}.\mathbf{k})$$

But $\mathbf{i}.\mathbf{j} = \mathbf{i}.\mathbf{k} = \mathbf{j}.\mathbf{i} = \mathbf{j}.\mathbf{k} = \mathbf{k}.\mathbf{i} = \mathbf{k}.\mathbf{j} = 0$

and $\mathbf{i}.\mathbf{i} = \mathbf{j}.\mathbf{j} = \mathbf{k}.\mathbf{k} = 1$

$$\therefore \mathbf{F}.\mathbf{S} = F_xS_x + F_yS_y + F_zS_z \qquad \textbf{5.6}$$

Since work done is the dot product of two vectors it is therefore considered as a scalar quantity.

Example 5.1. A ladder 3.0m long and weighing 200N has its center of gravity 1.2m from the bottom. At its top end is a 50N weight. Calculate the work required to raise the loaded ladder from a horizontal position on the ground to a vertical position.

Solution

The work done (against gravity) consists of two parts.

(1) The work done to raise the center of gravity of the ladder to a height of 1.2m. (2) The work done to raise the weight at the end of the ladder through 3.0m.

The work done to raise the center of gravity of the ladder = 200N x 1.2m = 240 J.

The work done to raise the weight at the end of the ladder = 50N x 3.0 =150 J

Total work done = 240 + 150 = 390 J.

Special Cases:

Case (i). When $\theta = 90^0$, then $W = FS\cos 90^0 = 0$ (since $\cos 90^0 = 0$)

So, work done by a force is zero if the body is displaced in a direction perpendicular to the direction of the force.

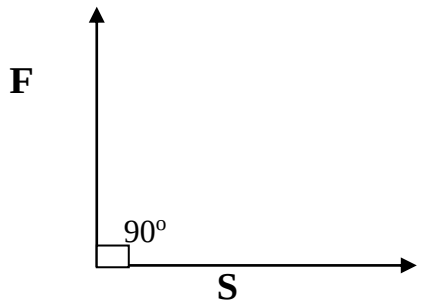


Figure 5.2

Example 5.2. Consider a body sliding over a smooth horizontal surface. The work done by the force of gravity and the reactional force of the surface will be zero. This is because both force of gravity and the reaction act normally to the displacement.

The same argument can be applied to a man carrying a load on his head and walking on a level street.

Example 5.3. Consider a body moving in a circle with constant speed. At every point on the circular path, the *centripetal force* and the *displacement* are mutually perpendicular. So the work done by the centripetal force is zero. The same argument can be applied to a satellite moving in a circular orbit. In this case, the gravitational force is always perpendicular to the displacement. So, work done by the gravitational force is zero.

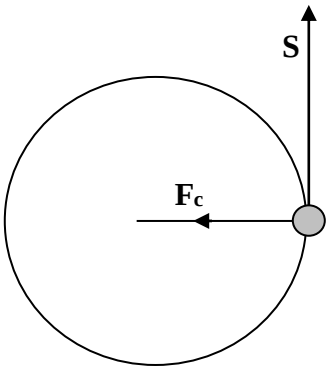


Figure 5.3 A body moving in a circular path

Example 5.4. The tension in the string of a simple pendulum is always perpendicular to the displacement. So, work done by the tension is zero.

Case (ii). When $\mathbf{S} = 0$, then $W = 0$.

So, work done by a force is zero if the body suffers no displacement on the application of a force.

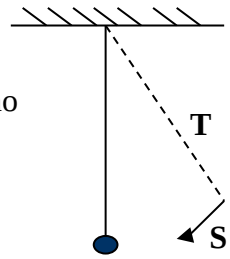


Figure 5.4 simple pendulum

Example 5.5: A man carrying a load on his head and standing at one place does no work.

Case (iii). When $0^0 \le \theta \le 90^0$, then $\cos\theta$ is positive .

Therefore, $W = FS\cos\theta$ is positive.

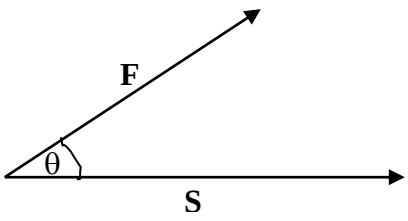


Figure 5.5

Definition: *Work done by a force is said to be positive if the applied force has a component in the direction of the displacement.*

Example 5.6. When a horse pulls a cart, the applied force and displacement are in the same direction. So work done by the horse is positive.

Example 5.7. When a body is lifted, the lifting force and the displacement act in the same direction. So work done is positive.

Example 5.8. When a spring is compressed, both the compressing force and the displacement act in the same direction therefore work done is positive.

Case iv. When $90^0 \leq \theta \leq 180^0$, then $\cos\theta$ is negative. Therefore $W = FS\cos\theta$ is negative.
Definition: *Work done by a force is said to be negative if the applied force has a component in a direction opposite to that of the displacement.*

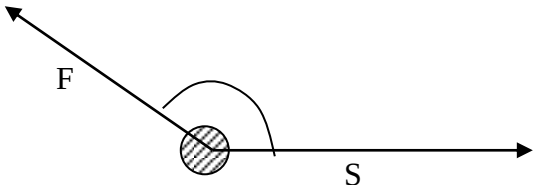


Figure 5.6

Example 5.9. When brakes are applied to a moving vehicle, the work done by the braking force is negative. This is because the braking force and the displacement act in opposite directions.

Example 5.10. When a body is dragged along a rough surface, the work done by the friction force is negative. This is because the frictional force acts in a direction opposite to that of the displacement.

Example 5.11. When a body is lifted, the work done by the gravitational force is negative. This is because the gravitational force acts vertically downwards while the displacement is in the vertical upward direction.

Example 5.12. A boy drags a bag of rice along a horizontal, smooth floor, with a force, F = 2N applied horizontally for a distance S = 3m. what work has the body done?

If the direction of F is not horizontal but an angle θ= 60° to the floor, what work has the boy done in this case?

Solution:

In the first case,

$W = FS = 2x3 = 6J$

In the second case,

$W = F\cos\theta\, x\, S = 2\cos 60^0 x3 = 3J$

Hence, for a given force and displacement, maximum work is done when the force is applied in the direction of the displacement.

Work done by a variable force

When the applied force varies both in magnitude and direction, then the total work done by the variable force is calculated by integration as described below.

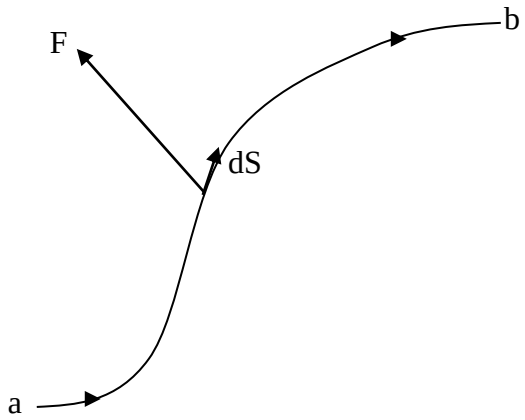


Figure 5.7 A path taken by a variable force

Suppose the variable force displaces a body from position a to position b along a curved path. Let us divide the path of the particle into a large number of infinitesimally small displacements. Each displacement should be so small that the force, during that displacement, may be regarded as constant.

Let dW be the small amount of work done in giving an infinitesimally small displacement $d\mathbf{s}$ to the body. Let $\mathbf{F}$ be the corresponding force. Then,

$$dW = \mathbf{F}.d\mathbf{s} \qquad \textbf{5.7}$$

The total work done W_{ab}, on the body as it moves from a to b can be obtained by adding up the elements of work done over each of the infinitesimally small displacements between a and b. This can be done by integrating both sides of Equation 5.7 to get

$$\int dW = \int \mathbf{F}.d\mathbf{s}$$

or

$$W_{ab} = \int F ds \cos\theta \qquad \textbf{5.8}$$

or

$$W_{ab} = \int (F\cos\theta) ds \qquad \textbf{5.9}$$

Equation 5.9 is the required expression for work done by a variable force.

Units of Work

Unit work is the amount of work done when a unit force displaces a body through a unit distance in the direction of the force. So, the units of work will depend upon the units of force and distance.

Absolute units of work. Work done is said to be *one absolute unit if an absolute unit of force moves a body through unit distance in the direction of the force*.

In SI units, the absolute unit of work is joule (abbreviated as J).

One Joule of work is said to be done *when a force of one Newton displaces a body through one metre in its own direction.*

$$1\,joule = 1newton \times 1metre$$

$$= 1kg \times 1ms^{-2} \times 1m$$

$$= 1kgm^2s^{-2} \text{ or } Nm. \qquad \textbf{5.10}$$

Energy

It is *the capacity or ability of a body to do work.*

It is a scalar quantity with unit and dimension is the same as that of work.

Energy is of many types: - mechanical energy, sound energy, heat energy, light energy, chemical energy, atomic energy, nuclear energy etc. The mechanical energy is of two types –kinetic energy and potential energy.

Kinetic energy is *the energy possessed by a body by virtue of its motion*. A body moving with high speed possess more kinetic energy than another body moving with lower speed.

Below are a few examples of bodies possessing kinetic energy.

(i) A ball in motion

(ii) A moving vehicle

(iii) Running water

(iv) Air in motion

Deriving expression for kinetic energy

Method 1.

Consider a body of mass m lying on a smooth horizontal surface as shown in Figure 5.8**.** Let a constant force **F** displace this body in its own direction through **S**. Let the velocity of the body, after traveling the distance **S** be **v**.

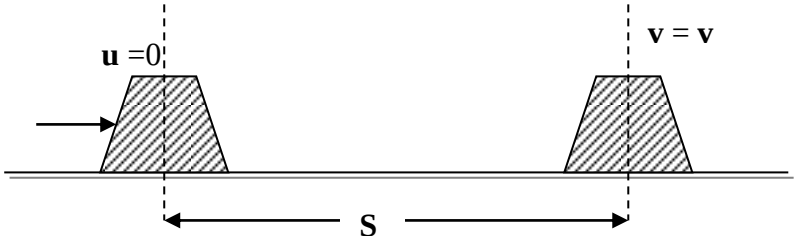


Figure 5.8 A force F displaces a body of mass m through a distance S

If W is the work done by the force, then

$$W = \mathbf{F}.\mathbf{S} = FS\cos\theta \qquad \textbf{5.11}$$

Both the force and the displacement are in the same direction

$$\Rightarrow \quad \theta = 0$$

Therefore,

$$W = FS\cos 0 = FS \qquad \textbf{5.12}$$

If **a** is the acceleration produced in the body, then

$$\mathbf{a} = \frac{\mathbf{F}}{m} \qquad \textbf{5.13}$$

From $v^2 - u^2 = 2aS$, with $u = 0$

$v^2 - 0^2 = 2aS$ so that

$v^2 = 2\left(\frac{F}{m}\right)S$ and using Equation 5.12

$$FS = W = \frac{1}{2}mv^2 \qquad \textbf{5.14}$$

This work done becomes the kinetic energy of the body. So, the kinetic energy of a body of mass m moving with velocity v is $W = \frac{1}{2}mv^2$

Calculus method.

If dW is the small work done in giving an infinitesimally small displacement $d\mathbf{s}$ to the body, then

$dW = \mathbf{F}.d\mathbf{s}$

Here, **F** is the applied force. Suppose the force and the displacement have the same direction, then,

$$dW = \mathbf{F}.d\mathbf{s} = Fds\cos 0 \qquad \textbf{5.15}$$

Let m be the mass of the body and **a** its acceleration. Then $\mathbf{F} = m\mathbf{a}$ and

$$dW = (ma)ds = m\frac{dv}{dt}ds$$

$$dW = m\frac{ds}{dt}dv = mvdv \qquad \textbf{5.16}$$

Let W be the total work done in increasing the velocity from zero to **v.** Then,

$$dW = \int_0^v mvdv = m\int_0^v vdv = m\left[\frac{v^2}{2}\right]_0^v$$

$$W = m\left(\frac{v^2}{2} - 0\right) = \frac{1}{2}mv^2 \qquad \textbf{5.17}$$

This work done becomes the kinetic energy. The kinetic energy of a body is one half the product of the mass of the body and the square of its speed. It is directly proportional to the (**i**) mass m of the body and (**ii**) square of the velocity v of the moving body.

The velocity **v** may be acquired by the body in any manner. The kinetic energy of a group of particles or bodies is the algebraic sum of the kinetic energies of the individual particles. Consider a system consisting of n particles of masses $m_1, m_2, ...m_n$. Let $v_1, v_2, ...v_n$ be their respective velocities. Then the total kinetic energy E_k of the system is given by

$$E_k = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 + ... + \frac{1}{2}m_n v_n^2 \qquad \textbf{5.18}$$

If m is measured in kg and v in ms^{-1}, then the kinetic energy is measured in joule. It may be noted that the units of kinetic energy are the same as those of work. In fact, this is true of all forms of energy since they are interconvertible.

Work-Energy Theorem

We know that $dW = mvdv$. If W is the work done in increasing the velocity from $v(0)$ to v, then

$$dW = \int_0^v mvdv = m\int_0^v vdv = m\left[\frac{v^2}{2}\right]_0^v$$

$$W = m\left(\frac{v^2}{2} - \frac{0^2}{2}\right) = \frac{1}{2}mv^2 - \frac{1}{2}mo^2 \qquad \textbf{5.19}$$

So, the work done on a body by a resultant force is equal to the increase or change in the kinetic energy of the body. This is called the Work-Energy Theorem or Work-Energy Principle.

Discussion of Work-Energy Theorem

(i) If there is no change in the speed of a particle, there is no change in kinetic energy. So, work done by the resultant force is zero.

Example 5.13. When a particle moves with constant speed in a circle, there is no change in the kinetic energy of the particle. So, according to the work-energy principle, the work done by the centripetal force is zero.

(ii) If the kinetic energy of the body decreases, then the work done on the body is negative. In this case, force and displacement are oppositely directed i.e. the force opposes the motion of the body.

(iii) In the above discussion, it has been assumed that the work done by the force is effective only in changing the kinetic energy of the body . However, it should be clearly noted that the work done on a body may be stored in the body as potential energy.

Potential Energy

It is the energy possessed by a body by virtue of its position in a force field or by its configuration.

Potential energy is also known as mutual energy or energy of configuration. The energy of a system can be altered by the application of a force.

Basically there are three types of potential energy. They are

(i) gravitational potential energy

(ii) elastic potential energy

(iii) Electrostatic potential energy (Electromagnetic).

Examples of bodies possessing potential energy due to position

(i) water stored in a reservoir

(ii) lifted weight

(iii) object hanging in space

Examples of bodies possessing potential energy due to configuration

(i) The coiled spring of a watch or a door lock.

(ii) A stretched bow or an elastic material

(iii) Compressed or elongated spring.

Gravitational potential energy near the surface of earth

As we go higher above the surface of the earth, the value of acceleration due to gravity decreases. But for those heights which are very small as compared to the radius of the earth, the acceleration due to gravity g and hence the weight mg of a body can be regarded as constant

Consider a body of mass m lying on the surface of the earth at a place where the value of acceleration due to gravity is g. The weight mg of the body acts vertically downwards. In order to lift the body, a force mg is required to be applied in the upward direction. Let h be the height through which the body is lifted. Work done W is given as

$$W = mg \times h$$

This work done is stored in the body as gravitational energy. Therefore gravitational potential energy is mgh. Strictly speaking, mgh is the gravitational potential energy of earth-body system. The height h can be taken from any level. The chosen level has to be fixed and considered as a reference level. The potential energy at the reference level is taken to be zero, and can be varied.

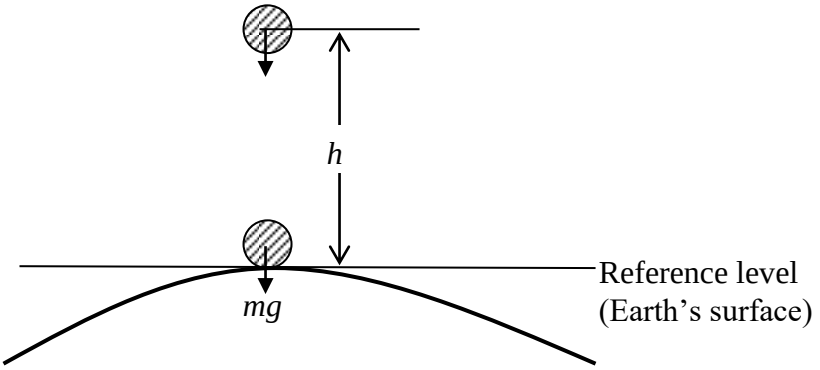


Figure 5.9 A body of mass m lying on the surface of the Earth has gravitational potential energy stored in it.

The potential energy at a point depends upon the reference level. However the difference of potential energies between two points is independent of reference level.

Conversion of gravitational potential energy to kinetic energy

When a body falls from a certain height, its kinetic energy begins to increase.

The gain in kinetic energy is at the expense of the gravitational potential energy. But the total mechanical energy remains conserved.

Consider a freely falling body of mass m to be at a position A. Let h be the height of the body above the reference level which is close to the ground.

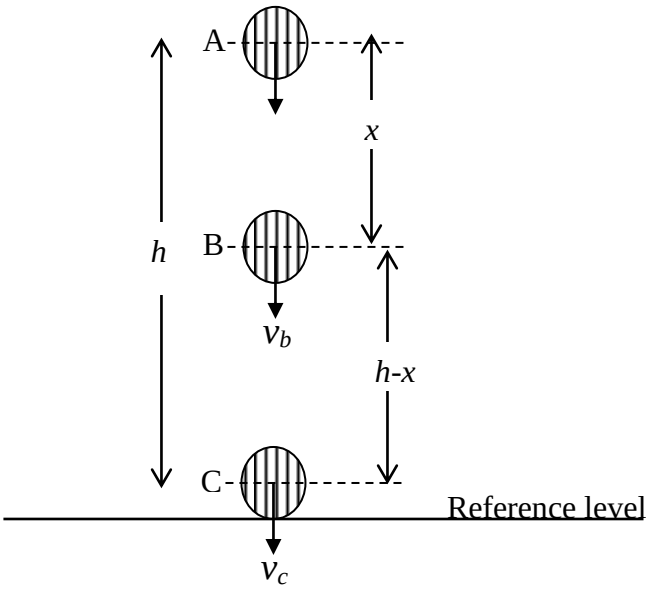


Figure 5.10 A body falling freely from a height h above a reference level

At the position A

Kinetic energy of the body is zero (the body is stationary)

Potential energy of the body is mgh

Total energy of the body is mgh

At position B

Let the body be at the position B at any instant after having fallen through a distance x. Potential energy of the body at B is

$mg(h-x)$

If v_b is the velocity of the body at B, then from equation 3.18

$v_b^2 - 0^2 = 2gx \Rightarrow v_b^2 = 2gx$

Kinetic energy of the body becomes

$$\frac{1}{2}mv_b^2 = \frac{1}{2}m \times 2gx = mgx \qquad \textbf{5.20}$$

The total energy of the body becomes

$$mg(h-x) + mgx = mgh \qquad \textbf{5.21}$$

At position C

When the body is at position C (just before impact with the ground), its height h above the ground may be regarded as zero. The potential energy of the body at C is zero.

If $\mathbf{v}_c$ is the velocity with which the body just touches the ground, then

$$v_c^2 - 0^2 = 2gh \Rightarrow v_b^2 = 2gh$$

The kinetic energy of the body then is

$$\frac{1}{2}mv_c^2 = \frac{1}{2}m \times 2gh = mgh \qquad \textbf{5.22}$$

Total energy of the body becomes mgh .

From the above deductions it is clear that the sum of the kinetic and potential energies of a freely falling body is constant at all stages of its motion. As the body falls, its potential energy decreases while its kinetic energy increases. However, the total mechanical energy is conserved. (Here, we have assumed that zero air resistance and hence no friction loss of energy).

Example 5.14. A ball of mass 1.5kg is thrown vertically upwards from the ground with an initial velocity of 100ms^{-1}. Calculate (a) the potential energy at a point A, 50m above the ground. (b) the kinetic energy at the point A. [assume g = 10ms^{-2}]

Solution

Mass of ball, $m = 1.5kg$, initial velocity, $u = 100ms^{-1}$, height above ground, $h = 50m$.

(a) $PE = mgh = 1.5 \times 10 \times 50 = 750\ J$

(b) To calculate the KE we have to first find the velocity of the ball at point A.

From kinematic equation of motion,

$v^2 = u^2 + 2gh$, *but* $g = -10ms^{-2}$ (motion against gravity)

$= 100^2 + [2 \times (-10) \times 50] = 9000ms^{-1}$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}x1.5x9000 = 6750J$$

Or $\frac{1}{2}mu^2 = p.e.(A) + k.e.(A) = mgh + k.e.(A)$ (Since p.e. + k.e. = constant)

$$k.e.(A) = \frac{1}{2}mu^2 - mgh = 7500 - 750 = 6750J$$

Example 5.15. A body of mass 1000kg is released from a height of 10m. With what energy does it strike the ground? [Take g=10ms^{-2}]

Solution:

$$E = mgh = 1000x10x10 = 1x10^5\ J$$

It strikes the ground with energy of 10^5J.

Transformation of energy in vibrating simple pendulum

The conversion of energy from one form to another is called transformation of energy. In Figure 5.11 a vibrating simple pendulum is shown. O is the point of suspension. When the centre of gravity of the bob is vertically below the point of suspension, the position of the pendulum is called the rest position or mean position or equilibrium position. Suppose the bob is displaced to the position E_1, and then released from this position, the pendulum will start vibrating between the two extreme positions E_1 and E_2.

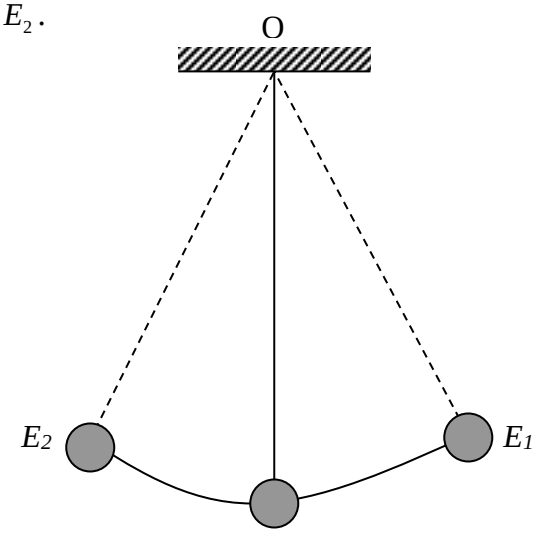


Figure 5.11 A vibrating simple pendulum

When the bob moves from the mean position to the extreme position E_1, its centre of gravity is raised vertically through a height h.

The potential energy of the body at the extreme position is mgh, where m is the mass of the bob and g is the value of the acceleration due to gravity at the place where the pendulum is suspended.

As the pendulum moves from the extreme position to the mean position, its potential energy goes on decreasing while the kinetic energy goes on increasing. However, the sum of the two energies is constant. At the mean position, the whole of the potential energy is converted into kinetic energy. If $\mathbf{v}$ is the velocity of the bob at the mean position, then the Kinetic energy at the mean position is

$$\frac{1}{2}mv^2$$

Applying the law of conservation of energy,

$$mgh = \frac{1}{2}mv^2 \text{ or } v^2 = 2gh \qquad \textbf{5.23}$$

Practical application of conversion of gravitational potential energy into kinetic energy.

The conversion of gravitational potential energy into kinetic energy has many practical applications. The most important application is 'hydroelectric power generation'. A large amount of water is stored in dams which are built at high levels. The water stored in the dam possesses a very large

amount of gravitational potential energy. When this water is made to run in the pipes, the gravitational potential energy is converted into kinetic energy. Now the fast-moving water is employed to run the turbines of the electric generators. This produces electrical energy.

Power

Power is defined as *the rate of doing work.*

We consider the power of an agent doing work if the time taken to complete a given amount of work is important to us.

The average power P delivered by any working device is given by

$$P = \frac{workdone}{time} = \frac{W}{t} \qquad \textbf{5.24}$$

W is the work done in time t.

But $W = \mathbf{F}.\mathbf{S}$ and velocity, $v = \mathbf{S}/t$

$$P = \frac{W}{t} = \frac{\mathbf{F}.\mathbf{S}}{t} = \mathbf{F}.\frac{\mathbf{S}}{t} = \mathbf{F}.\mathbf{v} \qquad \textbf{5.25}$$

From the definition of dot product, we can write

$$P = \mathbf{F}.\mathbf{v} = Fv\cos\theta \qquad \textbf{5.26}$$

where θ is the angle between $\mathbf{F}$ and $\mathbf{v}$.

If both $\mathbf{F}$ and $\mathbf{v}$ are in the same direction, then $\theta = 0$ and

$$P = \mathbf{F}.\mathbf{v} = Fv\cos 0 = Fv \qquad \textbf{5.27}$$

Instantaneous Power, P is the power of a working device at any instant.

Suppose a machine does an infinitesimally small amount of work dW in an infinitesimally small time dt, then

$$P = \frac{dW}{dt} = \frac{\mathbf{F}.d\mathbf{s}}{dt} = \mathbf{F}.\frac{d\mathbf{s}}{dt} = \mathbf{F}.\mathbf{v} \qquad \textbf{5.28}$$

Therefore mechanical power of a working machine at any instant is equal to the dot product of the applied force and the velocity at that instant.

The unit of power is joule per second (J/s) or watts (W).The power of a machine is said to be *one watt if it does one joule of work in one second*.

Example 5.16. Find the work done in moving an object along a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by F = 4i – 3j +2k. Calculate the power dissipated if it took 4s to move the object to its final position.

Solution

Work done = (Force field) . (Displacement vector)

= (4i – 3j +2k.) . [(2i – j + 4k) – (3i +2j – k)]

= (4i – 3j +2k.) . (-i – 3j + 5k)

= -4 + 9 + 10 = 15J

$$Power = \frac{workdone}{time\,taken} = \frac{15}{4} = 3.75W$$

Example 5.17. Calculate the power of a pump which lifts 200kg of water through a vertical distance of 2m in 4s. [g=10ms^{-2}]

Solution:

$$Power = \frac{W}{t} = \frac{mgh}{t} = \frac{200x10x2}{4} = 1000W$$